Summary: Project Phases

1. Data Preprocessing

• **Goal:** Convert raw data into a clean and usable format for machine learning.
• **Key Actions:** Loaded nearly 7,000 records, handled missing values, corrected data types, cleaned inconsistent categories, and encoded categorical variables (e.g., "Yes"/"No" → 1/0).

2. Exploratory Data Analysis (EDA)

• **Goal:** Understand patterns and key drivers behind customer churn.
• **Key Actions**: Identified that ~26.5% of customers churn, confirming an imbalanced dataset. Found strong patterns such as higher churn among month-to-month contract users compared to long-term customers.

3. Feature Selection & Model Creation

• **Goal:** Select the most important features and build a reliable predictive model.
• **Key Actions**: Performed feature engineering (e.g., AvgMonthlySpend, NumServices) to improve model insights. Compared multiple feature selection methods and models using F1-score. Selected Logistic Regression as the final model due to its better generalization and consistent performance on unseen data.

4. Final Model & Deployment

• **Goal**: Convert the trained model into a usable prediction system.
• **Key Actions**: Saved the trained model, scaler, and feature selection pipeline using pickle. Built a prediction flow that accepts user input, applies preprocessing steps, and outputs churn prediction in real-time.

Takeaway & Conclusion

• **Business Impact:**
The model helps identify high-risk customers (e.g., month-to-month users), enabling targeted retention strategies such as personalized offers, which can reduce churn and improve customer retention.